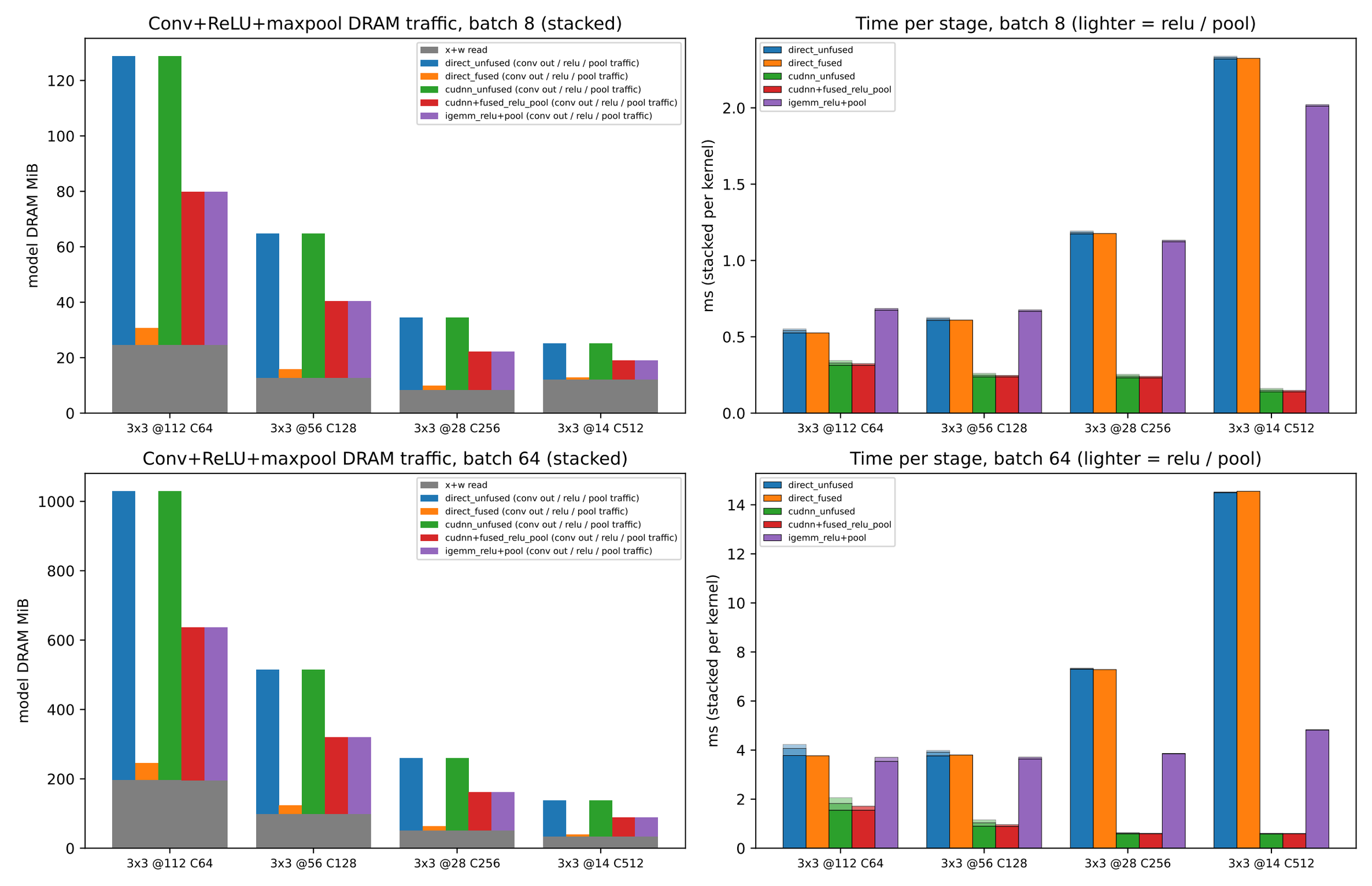
Conv+ReLU+maxpool DRAM traffic, batch 8 (stacked)
model DRAM MiB
x+w read
direct_unfused (conv out / relu / pool traffic)
direct_fused (conv out / relu / pool traffic)
cudnn_unfused (conv out / relu / pool traffic)
cudnn+fused_relu_pool (conv out / relu / pool traffic)
igemm_relu+pool (conv out / relu / pool traffic)
0
20
40
60
80
100
120
3x3 @112 C64
3x3 @56 C128
3x3 @28 C256
3x3 @14 C512
Time per stage, batch 8 (lighter = relu / pool)
ms (stacked per kernel)
direct_unfused
direct_fused
cudnn_unfused
cudnn+fused_relu_pool
igemm_relu+pool
0.0
0.5
1.0
1.5
2.0
3x3 @112 C64
3x3 @56 C128
3x3 @28 C256
3x3 @14 C512
Conv+ReLU+maxpool DRAM traffic, batch 64 (stacked)
model DRAM MiB
x+w read
direct_unfused (conv out / relu / pool traffic)
direct_fused (conv out / relu / pool traffic)
cudnn_unfused (conv out / relu / pool traffic)
cudnn+fused_relu_pool (conv out / relu / pool traffic)
igemm_relu+pool (conv out / relu / pool traffic)
0
200
400
600
800
1000
3x3 @112 C64
3x3 @56 C128
3x3 @28 C256
3x3 @14 C512
Time per stage, batch 64 (lighter = relu / pool)
ms (stacked per kernel)
direct_unfused
direct_fused
cudnn_unfused
cudnn+fused_relu_pool
igemm_relu+pool
0
2
4
6
8
10
12
14
3x3 @112 C64
3x3 @56 C128
3x3 @28 C256
3x3 @14 C512